

ABRAM S. ALFARO

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LinkedIn | GitHub

PROFESSIONAL PROFILE

Motivated and detail-oriented Magna Cum Laude graduate of Bachelor of Science in Computer Science from the Polytechnic University of the Philippines (PUP) with hands-on experience in software development through academic projects and a game development internship. Strong foundation in programming, database management, algorithm design, and web technologies, with proficiency in multiple programming languages and collaborative development tools. A consistent academic achiever and DOST-SEI RA 7687 scholar with strong problem-solving, teamwork, and leadership skills, eager to develop scalable, user-focused solutions and grow professionally in dynamic IT environments.

PROFESSIONAL EXPERIENCE

Game Developer Intern – Lamina Studios

Game Development Project | Video Output | Certificate of Completion | July — August 2025

Gained hands-on experience in game development by contributing to the design, programming, and testing of a horror video game project. Applied technical skills in Unity and C# to implement gameplay features, improve visuals, and enhance player experience.

- Developed gameplay mechanics (first-person controls, collectibles, triggers, AI enemy behavior) in Unity.
- Designed and implemented UI elements including main menu, pause menu, settings, objectives, and game-over screens.
- Enhanced visuals through lighting, post-processing, animations, and particle systems.
- Integrated audio effects and ambient sounds to improve player immersion.
- Assisted in scene sequencing, environment design, and final build testing for a fully playable horror game.

Programming Tools and Languages Used: C#, Unity

ACADEMIC PROJECTS

Hybrid PSO–ACO Logistics Optimization System | Tool and Thesis Manuscript | June 2025 — December 2025

Developed a hybrid optimization system combining Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) to improve 3D loading and unloading efficiency in delivery vehicles, addressing real-world logistics constraints such as space utilization and delivery sequencing.

- Designed and implemented a hybrid PSO–ACO algorithm to optimize both volume utilization and unloading efficiency by minimizing relocation count.
- Built a simulation-based web application to evaluate algorithm performance across varying package sizes and vehicle capacities using real-world datasets.
- Conducted comparative analysis against standalone PSO and ACO, measuring scalability through computation time and memory usage.

Course: Thesis Writing 1 & 2

Programming Tools and Language Used: Python, HTML, CSS

JARGEN | GitHub | Video Output | September 2024 — January 2025

Developed JARGEN, a personalized programming language for Generation Z Programmers. Inspired by Python and JavaScript, it takes advantage of the strengths of both languages while offering syntax catered towards Gen Z Coders.

- Programmed JARGEN's lexical and syntax analyzer with error detection for a smooth programming experience.
- Developed a web-based user interface for easy testing of code for JARGEN's lexical and syntax analyzer.

Course: Principles of Programming Languages

Programming Tools and Languages Used: Python, HTML, CSS

SSS Database | GitHub | Video Output | February 2024 — July 2024

Programmed a Database Management System (DBMS) for the Social Security System (SSS). The DBMS aimed to create a smooth registration process for SSS users and beneficiaries.

- Straightforward registration form interface with easy-to-understand input fields.
- Includes basic Create, Read, Update and Delete (CRUD) system design for database management connected to MySQL Workbench.

Course: Information Management

Programming Tools and Languages Used: MySQL Workbench, Java

BagEats | GitHub | Video Output | February 2024 — July 2024

Designed a Food Recommendation System for Polytechnic University of the Philippines (PUP) students titled BagEats. The recommender system is tailored towards the university's personnel making use of food data coming from actual food stalls found inside the PUP Main Campus.

- Implemented the Knapsack Algorithm as the backbone of the Food Recommendation System.
- Web-based user interface offering high customizability of food recommendations such as nutritional preferences and allergen avoidance.

Course: Design and Analysis of Algorithms

Programming Tools and Languages Used: Python, HTML, CSS

Bato-Bato Pick | GitHub | Video Output | June 2025

Developed a bato-bato pick (rock-paper-scissors) inspired card game with added power-up mechanics as a web application, enhancing traditional gameplay with strategy elements.

- Designed and implemented card-based power-up features (fire, water, lightning) to introduce variety and competitive depth.
- Built a web-based interface for interactive and user-friendly gameplay.
- Applied fundamental concepts of game logic and algorithm design to ensure fair and engaging matches.

Course: Automata and Language Theory

Programming Tools and Languages Used: JavaScript, HTML, CSS

OOPtify | GitHub | Video Output | January 2024

Developed a Java-based desktop application integrated with a Microsoft Access database containing Spotify data, designed to demonstrate object-oriented programming principles.

- Implemented OOP concepts such as inheritance, polymorphism, and encapsulation in the application's structure.
- Connected the system to a Microsoft Access database for storing, retrieving, and managing Spotify-related data.
- Designed a GUI in Java Swing to allow users to interact with the application easily.

Course: Object-Oriented Programming

Programming Tools and Languages Used: Java, Microsoft Access

EDUCATIONAL BACKGROUND

Polytechnic University of the Philippines (PUP) | Tertiary Education | 2022 — 2026

Bachelor of Science in Computer Science (BSCS) Graduate

- Graduated Magna Cum Laude | Proof of Latin Honor Award

Consistent President's Lister | 2022 — 2026

- GPA: 1.21

DOST-SEI RA 7687 Undergraduate Scholar | 2022 — 2026

Adamson University | Secondary Education – Senior High School | 2020 — 2022

STEM – Technology Strand Graduate, 2022

- Top Graduate of STEM – Technology Strand
- Graduated With High Honors

TOOL PROFICIENCY

Programming Languages: C, C#, Java, Python

Markup Languages: HTML, CSS

Database Management: SQL

Development Tools: Visual Studio Code, GitHub

Multimedia Tools: Figma, Canva, Adobe Photoshop, Adobe Premiere Pro

Productivity Tools: Microsoft Office (Excel, Word, PowerPoint), Google Workspace

SOFT SKILLS

- Leadership
- Creativity
- Teamwork
- Time Management
- Organized
- Critical Thinker